InfoCIR: Multimedia Analysis for Composed Image Retrieval

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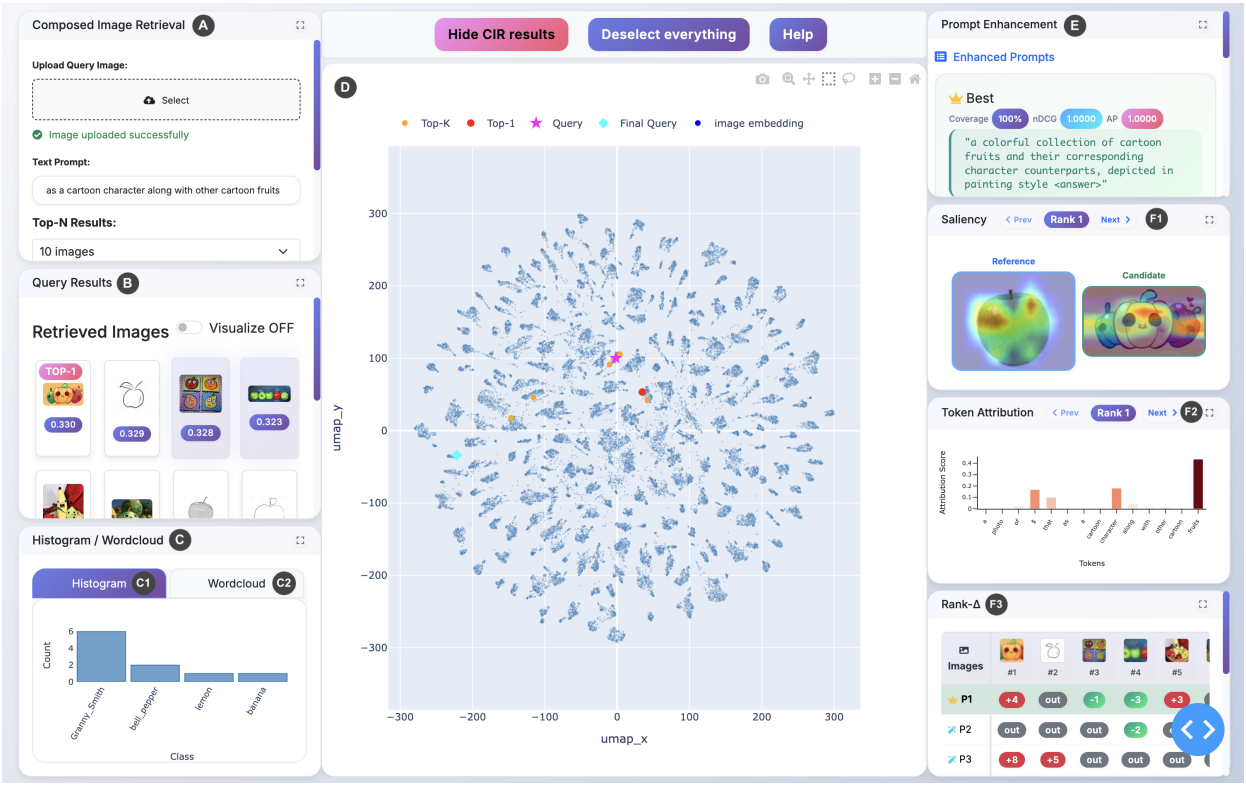

Fig. 1: The user interface of our system consists of six main panels. (A) The Composed Image Retrieval Panel allows users to input an image and a text prompt and select the number of images k to be retrieved. (B) The Query Results Panel displays the top-k images are listed by similarity; any image can be clicked to mark it as an ideal target for prompt refinement. (C) The Histogram / Wordcloud Panel includes class-frequency histogram (C_1) and a word cloud (C_2), summarizing labels within the current top-k. (D) The central Embedding View shows a 2D UMAP projection of the dataset, highlighting the reference image, the composed query embedding, and the top-k results. (E) The Prompt Enhancement Panel proposes alternative prompts conditioned on the selected ideals, using an LLM, and retrieval metrics. (F) The Explanation Panel visualizes model attribution using a saliency map (F_1), a token attribution bar chart (F_2) and a Rank-Δ heatmap (F_3).

Abstract—
Despite the growing popularity of CLIP-based Composed Image Retrieval (CIR), developers still lack tools that reveal how multimodal prompts interact with visual-textual embedding spaces and why small wording changes can dramatically alter the results. We present InfoCIR, a visual-analytics system that closes this gap by coupling retrieval, explainability, and prompt engineering in a single, interactive dashboard. InfoCIR integrates a state-of-the-art CIR back-end (SEARLE [1]) with a six-panel interface that (i) lets users compose image + text queries, (ii) projects the top-k results into a low dimensional space using Uniform Manifold Approximation and Projection (UMAP) for spatial reasoning, (iii) overlays similarity-based saliency maps and gradient-derived token-attribution bars for local explanation, and (iv) employs an LLM-powered prompt enhancer that generates counterfactual variants and visualizes their impact through an ideal-anchored Rank-Δ heatmap. A modular architecture built on Plotly-Dash allows new models, datasets, and attribution methods to be plugged in with minimal effort. We argue that InfoCIR helps diagnose retrieval failures, guides prompt enhancement, and accelerates insight generation during model development. All source code allowing for a reproducible demo, is available at `https://github.com/giannhskp/multimedia-analytics`.

Index Terms—Composed Image Retrieval, Prompt Enhancement, Saliency Maps, Token-level Attribution, Neural Ranking

1 Introduction

Composed Image Retrieval (CIR) is the task of retrieving images that resemble a reference image but differ in a specific way described by a natural language prompt [45]. Unlike traditional image search, which

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typically relies on either visual similarity or textual queries alone, CIR fuses both modalities into a single query. For example, a user may ask for images similar to a photo of a forest, but "in autumn colors" or "at night". Solving this task requires the system to interpret the transformation described in the prompt and apply it to the visual concept. CIR has gained attention in multimedia retrieval and computer vision [29, 37], especially with the introduction of vision-language models such as CLIP [31], which embeds both images and text into a joint feature space. These embeddings form the basis of many recent CIR models, including zero-shot systems that leverage pre-trained components without further task-specific training [5, 12, 53].

In practice, CIR remains difficult to work with. Developers and researchers often struggle to trace the causes of retrieval failures, compare model behavior, or refine prompts effectively. Moreover, small changes in wording can lead to dramatic shifts in output, and current systems offer little insight into why results appear as they do. These limitations hinder both system debugging and prompt engineering, especially in applications where precise control over image semantics is needed [38].

To address this gap, we present InfoCIR, a visual analytics system designed to support the exploration, diagnosis, and enhancement of CIR models through interactive embedding space analysis. InfoCIR brings together retrieval, attribution, and prompt enhancemnt in a single interface, enabling users to compose multimodal queries, compare models, interpret results, and refine prompts with immediate visual feedback.

As illustrated in Fig. 1, the interface integrates six linked components (A-F) that support interactive exploration. Users can select a reference image and a text prompt, view top-k retrieval results, inspect their distribution in the embedding space through Uniform Manifold Approximation and Projection (UMAP) [24], and identify key factors that influence model behavior. The system overlays saliency maps, token attributions, and prompt variants to help users understand and refine the search process. See the caption of Fig. 1 for a detailed breakdown of each component.

By letting users observe how retrieval results evolve as prompts are modified, InfoCIR supports a practical and interpretable framework for analyzing CIR model behavior. Users can track semantic drift and diagnose mismatches between intent and output. These capabilities support systematic failure analysis and informed prompt revision, contributing to more robust and explainable retrieval systems.

The core features of InfoCIR are designed to support explainability and interactivity. Our contributions include:

- After the initial image-plus-text query, users can mark one or more items in the top-k list as ideal. This explicit feedback anchors subsequent prompt enhancement, encouraging variants that generalize across several desired targets.
- An integrated explainable prompt enhancement module that (i) produces and compares LLM-generated prompt variants conditioned on the chosen ideals, (ii) visualizes their effect via an Ideal-anchored Rank-Δ heatmap, (iii) quantifies token impact using a gradient-based bar chart, and (iv) highlights spatial focus with saliency maps, supporting query enhancement, improved retrieval and human understanding of model behavior. Both saliency maps and token-attribution bars are produced by our in-house PyTorch re-implementation of Grad-ECLIP [59], refactored to work end-to-end with SEARLE.
- A metric-optimized UMAP projection places the top-k results, the reference image, and each composed query in a common 2D space, enabling users to inspect neighborhood structure, spot outliers, and assess how prompt changes move the query through the retrieval landscape.

Together, these features turn InfoCIR into a practical tool for diagnosing retrieval failures, understanding model behavior, and refining CIR workflows in an interpretable and interactive manner.

2 Related Work

This section reviews five research areas that shape the design and functionality of our system: (1) **Visual analytics for retrieval systems** provide interactive tools for examining ranking dynamics, analyzing query effects and refining results. (2) **Dimensionality reduction and embedding visualization** projects high-dimensional embeddings into navigable spaces, allowing users to inspect structural patterns and model behavior. (3) **Visual saliency for retrieval explanation** highlights image regions and textual cues that most influence similarity scores, revealing the model's decision process. (4) **Prompt enhancement and token attribution** link ranking changes to individual words and offer targeted, interpretable suggestions for query refinement. (5) **Positioning of our work** shows how the system unifies and extends these strands to deliver an explainable and interactive CIR workflow.

2.1 Visual Analytics for Retrieval Systems

Recent work in retrieval has increasingly emphasized interactive visual analytics to reveal ranking behavior, query formulation effects, and feedback loops. Early systems such as RankExplorer [35] visualized how rankings evolve over time, while Colorslope [46] extended this concept with slope graphs and heatmaps to capture dense ranking trajectories. These tools laid the foundation for systems that help users interpret the dynamic behavior of rankers in response to changing queries.

Some systems have explored prompt-based interaction at a higher level. PromptMagician [13] supports keyword recommendation and multilevel prompt editing for text-to-image generation, offering a practical interface for experimenting with compositional queries.

Another important direction is counterfactual analysis. CREDENCE [33] introduces query and document edits to explain ranking shifts, extended by recent frameworks for editable search explanations [6, 52]. In image retrieval, even a single well-chosen keyword has been shown to improve CLIP-based results significantly, without requiring retraining [25].

These developments illustrate a shift away from static evaluation toward interactive systems that help users analyze ranking dynamics, explore prompt effects, and understand retrieval behavior-principles that our system adopts and extends.

2.2 Dimensionality Reduction and Embedding Visualization

Visualizing high-dimensional data in a meaningful way requires dimensionality reduction techniques that preserve the structure of the original space. Classical methods such as Principal Component Analysis (PCA) [19] and Isometric Mapping (Isomap) [39] reduce dimensionality by preserving variance or geodesic distances, respectively. More recent methods such as t-distributed Stochastic Neighbor Embedding (t-SNE) and UMAP emphasize local neighborhood preservation and scalability, making them suitable for large-scale vision-language embeddings [24,42]. Other techniques including Triplet Map (TriMap) [3], which uses triplet constraints, and Pairwise Controlled Manifold Approximation and Projection (PaCMAP) [48], which balances global and local structure through staged optimization, have expanded the design space for projection-based analysis. These approaches have been applied in multimedia tasks, ranging from interactive music exploration [40] and fast image annotation [23], to clustering validation in dense text retrieval [22]. In our system, we adopt this strategy by using a fine-tuned UMAP projection to visualize the embedding space in a single, interpretable view that supports retrieval-driven diagnostics.

Building on these dimensionality reduction methods, embedding visualization tools allow users to explore and compare model representations interactively. TensorBoard's Embedding Projector introduced interactive PCA, t-SNE and UMAP plots with brushing and label coloring, providing one of the first practical interfaces for high-dimensional embedding analysis [36]. EmbeddingVis expanded this functionality to support side-by-side model comparison [21], while WizMap scaled these interactions to millions of points using quadtree summaries and map-style navigation [49]. Despite their utility, most of these tools remain model-agnostic and task-agnostic, treating each embedding as an isolated point. They lack retrieval-specific signals such as rank position, prompt evolution, and ideal-image anchors, all of which are integrated into our system [18, 58].

2.3 Visual Saliency for Retrieval Explanation

Visual saliency helps users interpret why a retrieval model ranks certain images as relevant, by highlighting the most influential regions or features. Similarity-based methods such as [10] perturb candidate images to identify areas that drive similarity scores, while interactive frameworks like [44] integrate these maps into feedback loops that enable real-time prompt or crop adjustments. Gradient-based techniques extend saliency to multimodal retrieval: Grad-ECLIP [59] adapts Grad-CAM [34] to CLIP by back-propagating through both image and text towers to produce joint heatmaps and token-level gradients, and SANE [28] links patch-level saliency to semantic attributes for improved interpretability. In the context of CIR, where queries combine visual and textual cues, such techniques help users understand how models align the modalities. Our system incorporates and adapts Grad-ECLIP to support post-retrieval explanation: after each search, a heatmap is overlaid on the reference or any retrieved image, connecting salient regions to influential prompt tokens and guiding targeted prompt enhancement.

2.4 Prompt Enhancement and Token Attribution

Early counterfactual studies showed that small query edits can reveal why certain items appear at incorrect ranks [6]. PromptBench [60] and Counterfact [8] extend this idea to language models by attributing rank shifts to individual tokens instead of entire prompts.

Token-level Gradient Attribution For multimodal models, gradient-based methods like Grad-ECLIP, SANE, and PromptExp compute signed influence scores for each sub-word across CLIP, image-text, and LLM tasks respectively [11, 28, 59]. These studies show that token-level signals more effectively reveal issues such as hallucinated attributes, compared to prompt-level saliency.

Visual summarization of edits While RankExplorer [35] and Colorslope [46] visualize rank shifts under prompt edits, they do not link changes to token-level causes. We adopt their color-encoded Rank-Δ design, but couple every row with a bar chart of per-token gradients, closing the causality loop between what moved and which word caused the move.

By combining token attribution with rank-change visualization, our system extends explanation techniques originally developed for text-only models to the multimodal setting of CIR. This integration bridges counterfactual query editing with saliency-based interpretation, enabling users to trace ranking changes back to specific words and their corresponding visual evidence.

2.5 Positioning of our work

Existing CIR models [1, 12] often focus on isolated components, overlooking the need for an integrated approach. Visual embedding tools typically ignore retrieval context and prompt semantics, while ranking interfaces emphasize accuracy without offering interpretability. Dimensionality reduction methods remain model-agnostic, providing limited insight into retrieval behavior, and prompt enhancement tools usually support only a single modality with minimal explanation.

Our system addresses these gaps by unifying retrieval ranks, prompt evolution, and token attribution in a single analytics platform, enabling direct comparison across CIR models. To our knowledge, it is the first system to jointly support analysis of embedding space, query dynamics, and model behavior.

3 Problem Definition & Design Goals

3.1 Formal Task

Composed Image Retrieval (CIR) targets the problem of retrieving images that resemble a given reference image but differ according to a textual modification. Unlike traditional image retrieval, CIR requires the system to interpret a natural language transformation and apply it to the visual input, fusing both modalities into a single, semantically meaningful query.

Formally, given:

- a reference image $I_{\text{ref}} \in \mathcal{D}$ (query image),
- a natural language modifier T (relative caption),
- a database $\mathcal{D}$ of images with precomputed embeddings,

the goal is to retrieve a ranked list of target images that are similar to I_{ref}, except for the modifications described by T. This is operationalized as a nearest-neighbor search in CLIP embedding space:

$$\big(I_{(1)}, \ldots, I_{(k)}\big) \;=\; \underset{I_j \in \mathcal{D}}{\arg\operatorname{sort}}\; \operatorname{sim}\Big(f_c(I_{\text{ref}}, T), f_i(I_j)\Big),$$

where f_c is a composition function that fuses the image and text into a query embedding, and f_i is a fixed encoder for the database images. In InfoCIR, we instantiate f_c using the state-of-the-art zero-shot model SEARLE and f_i using CLIP.

Prompt enhancement is the task of rewriting the original text modifier T so that the composed query embeds more accurately the user's intent. After the initial search, the user selects one or more ideal images I^* from the top-k results. A language-model-based editor $\mathcal{R}$ then produces a small set of alternative prompts conditioned on the visual context of the reference image I_{ref}, the original text modifier T, and the content of the chosen ideal image(s):

$$T' = \mathcal{R}\big(I_{\text{ref}},\, T,\, I^*\big).$$

This procedure is executed once per user selection, and no multi-turn refinement loop is applied. Each candidate prompt T' is re-evaluated by the retrieval engine, and its effect on the rank of I^* is displayed in the Ideal-anchored Rank-Δ heatmap, alongside token- and pixel-level attribution visualizations. Because the backbone embeddings remain fixed, this single-pass rewrite can boost retrieval quality without any model retraining.

3.2 Design Goals

Building on our analysis of CIR workflows, interaction patterns, and retrieval behavior (see Sections 2 and 5), we define four complementary design goals that guide every interface and algorithmic choice:

DG1 Assess Enable rapid judgment of retrieval quality. Provide visual context through UMAP-based embedding maps, local similarity signals, and evidence overlays such as saliency maps, so users can spot failure modes at a glance.

DG2 Compare Support direct comparison across prompt variants and retrieved images. Linked views must visualize rank changes, embedding drift, and class balance instantly when users toggle between configurations.

DG3 Enhance Close the loop with actionable edits. A prompt enhancement module proposes minimal token edits, the Rank-Δ heatmap reveals whether the ideal image rises in rank, and attribution feedback indicates which words contribute most.

DG4 Understand Expose underlying causes. Token attribution and similarity-based saliency link language to visual content, helping users build transferable mental models of retrieval behavior.

Together, these design goals enable a diagnose-compare-enhance cycle that mirrors how users explore, evaluate, and improve composed image queries. This cycle underpins the interface and methodological contributions described next.

4 Methodology

The proposed interactive multimedia analytics framework for Composed Image Retrieval (CIR), combines multimodal deep learning with coordinated visual representations to support prompt refinement and model introspection. It centers around user feedback, interpretable visualizations and prompt enhancement strategies, leveraging foundation models. The system follows a modular pipeline architecture, shown in Fig. 2, inspired by the work of Worring et al. [51], with distinct stages for input, model selection, retrieval, projection, visualization and prompt enhancement. The methodological components of the system are described below.

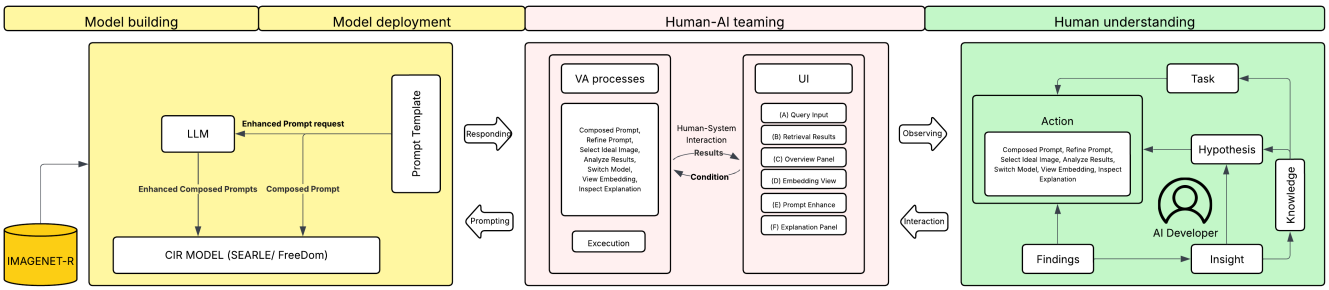

Fig. 2: The architecture links prompt composition, embedding-based retrieval, and interactive visualization across UI panels (A-F), enabling users to enhance prompts, interpret outputs, and derive insights. Following Wang et al. [47], we integrate semantic fusion (CLIP space), visual integration (saliency and token overlays), and cross-view linking (synchronized views).

4.1 System Overview

The pipeline begins with a user-provided input query consisting of a reference image and an accompanying natural language prompt. It integrates the SEARLE [1] CIR model, which uses CLIP to compute joint image-text embeddings for comparative retrieval and visualization workflows.

These embeddings are used to rank images in a target database by similarity. The top-k most similar results are visualized using a suite of coordinated views including a 2D scatterplot, image gallery, class distribution histogram, and semantic word cloud. An integrated explainable prompt enhancement module uses an LLM to generate alternative text variants, re-ranks them against a user-chosen ideal image, and instantly visualizes their impact through saliency maps, compact Rank-Δ heatmaps and token-influence summaries, giving users a rapid, transparent loop for refining their queries.

The system is implemented using the Plotly and Dash frameworks, which support highly interactive, web-based visualizations with seamless integration of Python-based machine learning components, as illustrated in the system overview image (Fig. 1).

4.2 Input and Model Selection

The system accepts two primary inputs: (1) a reference image which is uploaded and (2) a textual prompt describing the desired transformation or characteristics. A third, optional (but strongly recommended) input is the ideal image(s): from the initial top-k list the user selects one or more images that align with their intent. These images, become the fixed anchor, against which all prompt variants are evaluated, enabling fine-grained counterfactual diagnostics described in Sec. 4.6.

Users employ the SEARLE CIR back-end, a zero-shot retrieval model that performs textual inversion. The reference image is transformed into a pseudo-token within CLIP's text embedding space, allowing compositional fusion with the user's prompt.

The model projects the image and text into a shared multimodal embedding space (i.e., CLIP space), producing a joint query vector. This embedding is used to compute cosine similarities against pre-computed image embeddings in the database.

4.3 Retrieval and Similarity Computation

Given the composed query embedding, the system ranks all database images using cosine similarity. The top-k images are selected for further visualization, forming a ranked result set. Alongside the images, similarity scores and, when available, associated class labels or tags are returned for further analysis.

4.4 Dimensionality Reduction

To visualize the retrieval space, we pre-train a metric-optimized UMAP on the full corpus of image embeddings. This offline training stage yields a low-dimensional manifold that attempts to balance local fidelity (trustworthiness) with global layout stability (continuity). At run time, the top-k retrieved images are simply looked up in this pre-computed map, ensuring that every point occupies a position that is globally consistent with the entire dataset.

Crucially, UMAP's out-of sample transformation allows us to embed new vectors such as the reference image, the composed query embedding $f_c(I_{\text{ref}}, T)$, and any subsequent query embeddings generated after prompt enhancement into the same 2D coordinate system.

This capability keeps all queries and their evolving results in a single geometric frame of reference, allowing users to observe how different prompts shift the query's position relative to the broader semantic landscape.

4.4.1 Fine-Tuning the UMAP Embedding Space

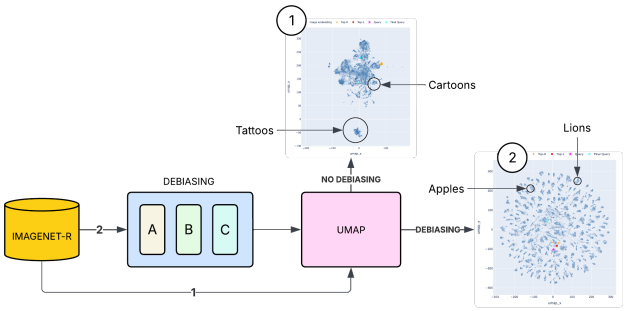

Fig. 3: Conceptual overview of the UMAP fine-tuning pipeline. ImageNet-R embeddings are passed through a three-stage debiasing process, (A) Style Debiasing, (B) Contrastive Debiasing, and (C) Alternative Projection before UMAP projection. Compared to the unprocessed projection (1), the debiased version (2) exhibits improved semantic clustering and reduced influence of style artifacts.

The UMAP fine-tuning pipeline, illustrated in Figure 3, is designed to improve the dimensionality reduction and visualization of CLIP image embeddings, with a particular focus on enhancing semantic clustering and eliminating style-based artifacts. This is achieved through a multi-stage process that begins with standard CLIP embedding extraction, followed by a sophisticated debiasing and preprocessing pipeline. The debiasing step incorporates three techniques: (1) Style Debiasing, which first applies PCA and then filters to retain the top 50% of dimensions with the highest semantic discriminability to focus on semantic content, (2) Contrastive Debiasing [16], which moves each embedding 70% closer to its class prototype to enhance semantic consistency, and (3) Alternative Projection via Independent Component Analysis (ICA) [30], aimed at separating style and content features. Overall, the use of debiasing methods is intended to reduce the influence of style-based clustering in CLIP embeddings and to promote clustering based on class labels instead.

We fine-tune UMAP to improve both global layout and within-class cohesion.

Broader neighborhoods Enlarging the search radius captures more contextual information, yielding a smoother global layout and clearer inter-cluster separation. (`n_neighbors = 25, min_dist = 0.1, spread = 55`).

Deeper optimization. Extending the training to 1200 epochs lets the manifold converge more faithfully (`n_epochs = 1200`).

Label-guided clustering A light supervisory signal pulls samples toward their class centers, sharpening semantic clusters (`target_metric = categorical, target_weight = 0.3`).

Dense local links Raising local connectivity preserves fine-grained neighborhood relations, so subtle within-class variations remain visible (`local_connectivity = 3`).

Contrastive debiasing A pre-projection step with a high contrastive weight pulls embeddings toward class prototypes, suppressing style artifacts and boosting semantic distinctiveness before the final UMAP step (`contrastive_weight = 0.7`).

To validate the effectiveness of this enhanced pipeline, we compute a comprehensive suite of four quantitative quality metrics:

- **Trustworthiness**: measures how many points that are far in the high-dimensional space appear spuriously close after projection (ideal = 1).
- **Continuity**: assesses how many genuine near-neighbors in the high-dimensional space remain neighbors after projection (ideal = 1).
- **Average Local Error**: the mean Euclidean distortion within the k-nearest neighborhood of every point (lower values indicate better local fidelity).
- **Neighborhood Hit**: a supervised score that counts how many of the k projected neighbors share the same ground-truth class label (higher values signal cleaner class separation).

In Table 1 of Appendix A we report the values of the first and the final UMAP fine-tuning steps.

4.5 Interactive Visualization

The retrieved and projected images are rendered in multiple coordinated visual views:

- **Scatterplot View:** Images are plotted based on their 2D embeddings. Proximity indicates semantic similarity, allowing users to observe clustering or outliers. Interactive selection reveals metadata on the respective projection.
- **Image Gallery:** Displays the retrieved images sorted by similarity to the query. This view supports rapid inspection of ranked outputs and is synchronized with the scatterplot for cross-reference.
- **Class Histogram:** Shows the frequency distribution of classes among the retrieved results, supporting analysis of category bias or dominance.
- **Word Cloud:** Aggregates tags, labels, or captions associated with the retrieved images. Frequent terms are emphasized, offering an overview of semantic trends in the result set.
- **Prompt Enhancement Panel:** Displays alternative prompts generated by an LLM, optimized to improve retrieval of one or more user-selected ideal target images. Users can compare variants interactively, with each selection triggering a full update of associated retrieval outputs and explanatory views.
- **Saliency Map View:** Overlays gradient-based visual saliency on the reference and retrieved images. This map highlights which image regions most influence similarity scores, linking visual evidence to textual input and assisting users in understanding model focus.
- **Token Attribution Chart:** Provides token-attribution scores based on gradient analysis. It quantifies how strongly each token in the prompt contributes to the retrieval of the ideal image.
- **Rank-Δ Heatmap:** Visualizes how each prompt variant affects the standing of the originally retrieved images. Rows represent prompt variants and columns list the top-k images returned by the initial query in rank order. Cell colors show rank change relative to the baseline (green for upward moves, red for drops), allowing users to spot at a glance which variants meaningfully boost (or harm) the ranking of their targets.

These views are linked, in order to support multi-faceted exploration. Interactions in one view update the others.

4.6 Explainable Prompt Enhancement

The prompt enhancement module in InfoCIR is a user-guided process rather than an automated iterative loop. After reviewing the initial retrieval results, the user selects one or more ideal target images from the top-k. These chosen images serve as anchors for enhancement: the system leverages a large language model (LLM) to suggest a set of semantically enriched prompt variants grounded on the content of the ideal images. For each proposed variant, the system re-ranks the results and visualizes the ideal image's rank change via an Ideal-anchored Rank-Δ heatmap. This heatmap provides immediate feedback on how each new prompt affects the ideal image's position in the results, helping users compare prompt efficacy at a glance.

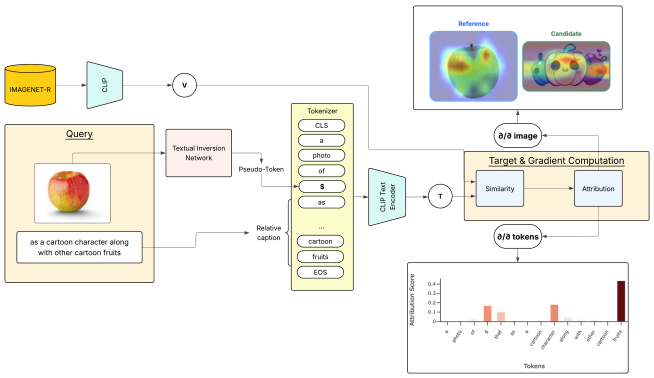

Fig. 4: Explanation pipeline for visual and textual attribution in CIR. The system accepts a reference image I_r and a relative caption C, combining them into the prompt "a photo of $\varphi(I_r)$ that C", where $\varphi(I_r)$ is a pseudo-token generated via a textual inversion network. This prompt is tokenized and encoded by CLIP's text encoder to produce a text embedding T. Candidate images from the dataset (e.g., ImageNet-R) are encoded into image embeddings V, and retrieval is based on cosine similarity $s = \cos(V, T)$. To generate explanations, the similarity score s is differentiated with respect to either the image or the text tokens. The resulting gradients yield visual saliency maps (top right) and token attribution scores (bottom right), helping users understand which parts of the image or prompt influenced the retrieval result most.

Along with the prompt suggestions, the module provides explainability tools to highlight why certain prompts perform better. A token attribution chart (using gradient-based analysis [11]) highlights which words in the text prompt have the strongest impact on the ideal image's retrieval score. Technically, we compute the gradient of the similarity score between the composed query and a retrieved image with respect to each input token. The magnitude of these token-level gradients indicates each word's contribution to the match. The system visualizes this as a bar chart, surfacing high-impact tokens and thereby guiding the user toward terms that most influence the model's decision. This offers insight into the model's reasoning about the text query and suggests which keywords might be emphasized or adjusted in further refinements (see Fig. 4).

Complementing the token-level explanation, a saliency map is generated to show which regions of the reference image are most influential for retrieval relevance. We employ gradient-based visual explanation

techniques, and more specifically implement from scratch an adaptation of Grad-ECLIP [59] for our CLIP-based multimodal encoder (SEARLE), to attribute the composed query embedding's output back to the image pixels. In practice, this involves back-propagating the retrieval score's gradient through the model's image tower, producing a heatmap overlay on the reference image that denotes areas crucial to the query-result similarity. The saliency highlights enable users to see which parts of the image the model "looked at" when matching the query, linking concrete visual evidence to the semantic query feedback (see Fig. 4).

Together, these explainable prompt enhancement features put the user in control of an interactive feedback loop. The LLM-generated prompt variants (with their rank-change indicators) and the attribution visualizations (token bars and saliency maps) collectively, allow users to steer the search with greater transparency and confidence. This approach aligns with emerging interactive explanation paradigms that emphasize human-in-the-loop refinement and deeper model transparency [33], ensuring that prompt adjustments are both informed by the model's inner workings and grounded on the user's intent.

4.7 Metadata

At each stage of the pipeline, quantitative signals are logged to support downstream analysis. These include: similarity scores for each retrieved result, the 2D projections of the embedding vectors, class labels, and user interactions (e.g., selected prompts).

This data can be used to evaluate model sensitivity, compare methods and support reproducibility.

4.8 System Modularity and Extensibility

The system is designed with modularity in mind. Retrieval models can be easily swapped or added as long as they provide image and text embeddings in the CLIP embedding space. Visualization modules are decoupled and configurable, supporting the addition of new views or encoding strategies. This flexibility allows for experimentation with retrieval algorithms, datasets and user interaction paradigms.

5 Interaction Design for CIR

Following Pike et al.'s "science of interaction" [27], we treat each user action in CIR as a micro-scale hypothesis test. A user poses an analytic goal (e.g., compare two prompts), acts on the interface, observes the system's response, and updates their mental model. This feedback loop supports interactive visual reasoning.

To enable high-level analytic goals (*explore*, *assess*, *compare*, *refine*, *validate*, *understand*, and *triage*) our design applies Yi et al.'s taxonomy of seven intent-driven interactions [55]: *Select, Explore, Reconfigure, Encode, Abstract/Elaborate, Filter*, and *Connect*. Each maps to a cognitive lever: marking (Select), navigating (Explore), switching models or prompts (Reconfigure), adjusting visuals (Encode), toggling detail (Abstract/Elaborate), narrowing focus (Filter), and coordinating views (Connect).

Users often chain these intents to perform broader tasks. For example, to compare prompts, they may: (1) **Select** a variant, (2) **Reconfigure** the result list, and (3) **Encode** by similarity, before assessing whether the change helped. The system supports this cycle with immediate visual feedback.

The following subsections map these high-level goals to specific CIR tasks and detail the low-level techniques that realize them in interaction.

5.1 High-level interaction intents

Following the cognitive sense-making stages described by Pike et al. [27], we characterize our high-level interaction goals (*Explore, Assess, Compare, Refine, Validate, Understand, and Triage*) as the user's macro-level objectives while analyzing CIR results.

- **Explore**: Navigate the UMAP map, pan–zoom the gallery, and skim class histograms to discover clusters, outliers, or previously unseen concepts.
- **Assess**: Judge retrieval quality by checking similarity scores, saliency overlays and token bars for the current top-k list; decide whether the result set is "good enough".
- **Compare**: Switch between different prompt variants to see which formulation ranks the ideal images higher and produces the most coherent neighborhood in the UMAP projection.
- **Refine**: Use the LLM prompt-enhancer to iterate on wording. Selecting a candidate prompt constitutes a single turn in the dialogical inquiry loop described by Pike et al., with the Rank-Δ heatmap updating live to test emerging hypotheses.
- **Validate**: Confirm that the chosen "ideal" image(s) climb the list under a new prompt; rely on the Rank-Δ heatmap's green upward shifts, the scatterplot's positional change and the extra insights visualized for each prompt.
- **Diagnose & Understand**: Toggle token-attribution bars and saliency maps to examine model reasoning; any unexpected change signals a helpful breakdown in interaction (Pike et al.) that can lead to deeper understanding or reformulated goals.
- **Triage**: Via the class histogram or word cloud, spot over-represented categories and decide whether to filter them out or re-weight the query focus.

5.2 Low-level interaction intents

Each high-level intent is enabled by fast, intuitive interaction techniques from [55]. These support fluid visual reasoning and immediate feedback.

- **Select / Click**: Single- or multi-click one or more image thumbnails to set them as ideal targets, or click an enhanced prompt variant to visualize its token attribution chart, saliency maps, the top-k results for this prompt in the scatterplot and the image gallery.
- **Lasso & Multi-select**: Drag a rectangular (lasso) region in the scatterplot to bulk-highlight a cluster; the histogram and word cloud update instantly.
- **Pan–Zoom**: Zoom and click-drag pan in the projection for macro- / micro-inspection; dedicated button recenters on the current focus.
- **Dropdown / Toggle**: Use the toggle switch to visualize the retrieved images in the scatterplot, or a dropdown to change k (result list length); toggles keep interaction latency below the "immediate" threshold.
- **Brushing & Linking**: A selection in any view (gallery, histogram, word cloud) instantly highlights counterparts everywhere else, sustaining the cross-view interpretability chain advocated by Pike et al.
- **Encode-on-Demand**: Apply color codes on the fly. Color by rank position, processing phase (query image vs. composed query), or current selection, so users can recognize patterns instantly without regenerating the layout.

5.3 Usage Scenario

To demonstrate how InfoCIR supports interactive refinement in CIR, consider a user uploading an image of a green apple and entering the prompt "a red apple". The goal is to retrieve semantically relevant images that preserve the object identity (apple) while modifying its attribute (color).

Upon querying, the SEARLE retriever returns a result set that includes both green and red apples, along with unrelated fruits. In the UMAP projection, results are scattered, with no clear cluster for red apples. The user opens the attribution panel, which shows that the term "red" contributes weakly to the ranking, suggesting a style bias in the CLIP representation.

To correct this, the user selects one or more retrieved images that match their intent (e.g., any clearly deep-red apple) and opens the

Prompt Enhancement module. This component suggests semantically guided prompt variants, such as "ripe apple", "crimson apple", and "dark red fruit", each grounded on corpus embeddings and visual similarity.

After choosing "crimson apple", the updated retrieval results appear in a compact cluster of red apples in the embedding space. The selected image jumps from rank 12 to rank 2, as shown in the Rank-Δ heatmap, and the attribution weights now reflect stronger activation on "red" and relevant image regions.

This scenario illustrates how InfoCIR enables users to debug and steer the retrieval process through prompt adaptation and embedding inspection, leading to semantically faithful results that align with the composed intent.

6 Application evaluation roadmap

The following roadmap outlines planned evaluation studies. None of those has been carried out yet, but they will be executed in the next development phase, using the alpha edition of our application InfoCIR (alpha), according to our original working plan.

Guided by visual analytics practices [50, 57], we will gather converging evidence from four complementary studies.

Expert heuristic audit Two visual analytics experts and one UX specialist will conduct a paired cognitive walk through. They will score violations of the dashboard design patterns, screenfit, overflow, brushing and linking, parameterization, and detail on demand [4, 14]. All critical issues must be resolved before any user study, ensuring that later evaluations measure analytic performance rather than basic usability defects.

Agent-based simulation Scripted oracle agents will solve three benchmark retrieval tasks. For each single-pass prompt enhancement, we record the number of user-interface actions as interaction cost C and count the additionally retrieved correct target images as knowledge gain K. We then compute the analytic-quality ratio K/C as proposed by van Wijk [43]. Values above 1.5 will be taken to indicate an acceptable cost-benefit trade-off, showing that improvements are not marginal, but deliver at least 50% more benefit than cost.

Controlled A-B testing study Twelve MSc students will complete four retrieval scenarios in a within-subjects design, counterbalanced between InfoCIR and a Tableau gallery baseline. We will record task completion time, mean reciprocal rank (MRR), and user feedback using the System Usability Scale (SUS) questionnaire [41], which measures perceived ease of use and trust. Each scenario will be capped at ten minutes. Because of the small sample size and the paired nature of the measurements, we will apply the Wilcoxon signed-rank test [32] to assess whether performance and usability differences between the two systems are statistically significant.

Insight-based case study Four domain experts, e.g., researchers or developers in Information Retrieval, will each perform a 45 minute think-aloud recorded exploration. Insights will be coded using the five-level hierarchy of North [26]. Success will be defined as each participant generating at least one hypothesis-level insight, with an overall average insight density exceeding 0.25 insights per minute (15 insights per hour), ensuring that participants will remain engaged and continually uncover findings.

Ethics All participants will provide informed consent prior to the study and will be informed of their right to withdraw at any time without penalty. No personally identifiable information will be collected, and all interaction logs will be anonymized and stored securely in compliance with institutional data protection guidelines.

These planned steps will move InfoCIR from proof of concept toward a validated tool for explainable composed image retrieval.

7 Discussion and Future Work

The development of InfoCIR highlights the growing need for explainable, interactive systems in the domain of Composed Imagex Retrieval (CIR). While recent CIR models show promising zero-shot and training-free retrieval capabilities [1, 12], their behavior often remains opaque to users, particularly when multimodal queries (image + text) yield unexpected results. Our system addresses this challenge by providing a tightly integrated workflow, through which users can trace retrieval dynamics through interpretable and visual means.

A key design insight was the benefit of combining retrieval results with both the overall structure of the embedding space and detailed feedback on individual prompt tokens. Embedding projections via UMAP support exploration of neighborhood coherence and cluster structure, while token attribution and saliency maps provide insights on the internal reasoning behind model decisions. The connected and interactive visual views allow users to detect and act on misalignments, such as when a prompt term appears dominant despite being semantically not relevant, making it easier to quickly form and test new ideas.

Another important contribution is the idea of ideal-guided prompt enhancement. By allowing users to fix one or more target images as benchmarks, InfoCIR turns prompt enhancement into a constrained optimization task visible through the Rank-Δ heatmap. This approach offers immediate, localized feedback on prompt effectiveness, addressing a longstanding gap in model debugging workflows, where prompt quality is typically judged only by final retrieval rank without insight into causal dynamics [7, 20].

However, limitations remain:

- Dimensionality reduction methods, while informative, can distort fine-grained relationships and may mislead interpretation if not carefully configured [15]. As future alternatives, we plan to explore PaCMAP, which is known for preserving both local and global structures through staged optimization, as well as TriMap, that emphasizes triplet-based constraints to maintain global geometry in projections [3, 48].
- Token attribution charts depend on gradient approximations that may not fully capture attention in transformer-based architectures [17].
- While prompt suggestions are LLM-generated, they remain heuristic and do not guarantee improvement, particularly in abstract or ambiguous queries. Future iterations could incorporate structured prompts (e.g., scene graphs [56] or template slots), multi-turn prompt refinement [54], and automatic failure detection [9] to guide LLM-based enhancements.
- Attribution-based explanations, while useful, offer only local insight. Extending the system to support counterfactual visual generation (e.g., what-if edits to the query image) or concept-level abstractions could provide global interpretability and support model debugging at a higher level [2].
- Current refinement consists of a single LLM-based rewrite guided only by the user-selected ideal images. Saliency maps and token-attribution bars serve purely as explanatory overlays and are not fed back into the rewrite function. Future iterations could let explicit user annotations, such as region sketches, custom attention masks, or reinforcement-guided scoring, directly influence the prompt generator, providing finer control and richer engagement.

Despite these constraints, InfoCIR demonstrates that aligning retrieval outcomes with human-centered reasoning is both feasible and impactful. The system advances CIR tooling beyond black-box metrics toward interactive, explanation-rich workflows that better support model development, evaluation and refinement.

8 Conclusion

We presented InfoCIR, a visual analytics system designed to enhance transparency, interpretability and refinement in Composed Image Retrieval (CIR). By integrating retrieval ranking, embedding projection and visualizations, prompt enhancement and multi-level explanation within a coordinated and interactive interface, our system addresses key gaps in CIR interpretability and supports developers in exploring, understanding and improving and validating retrieval model behavior with greater confidence.

Our contributions include an explainable prompt-enhancement pipeline that combines LLM-generated prompt variants with gradient-based attribution visualizations, a coordinated environment for inspect-

ing embedding dynamics, and a modular architecture that can accommodate new CIR models with minimal effort. Through guided interface interactions, users can rapidly assess prompt impact, explore the embedding landscape, and pinpoint failure cases or performance bottlenecks.

In contrast to existing tools that isolate embedding inspection or prompt engineering, InfoCIR combines both, supporting a new paradigm of retrieval-centered multimodal analysis.

Future work can extend model support, scalability and personalization, further bridging the gap between opaque model internals and user-centered retrieval workflows. As the field of multimodal AI continues to evolve, we see InfoCIR as both a diagnostic toolkit and a framework for future systems that prioritize human-guided refinement, transparency and trust in retrieval pipelines.

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A Fine-Tuning Results

Table 1: Comparison between initial and final configurations of the dimensionality reduction setup used in our UMAP-based visualization pipeline (Sec. 4.4.1). For `trustworthiness`, `continuity`, and `neighborhood_hit`, higher values indicate better quality. For `average_local_error`, lower values are better. Bold values highlight the best performance across each metric.

Parameter / Metric	Initial	Final
`n_neighbors`	15	25
`min_dist`	0.1	0.1
`spread`	1.0	55.0
`metric`	cosine	cosine
`pca_components`	–	80
`n_components`	2	2
`target_weight`	–	30%
`target_metric`	–	categorical
`contrastive_weight`	–	70%
`local_connectivity`	–	3.0
`n_epochs`	200	1200
`n_classes`	200	200
`n_samples`	30000	30000
`trustworthiness`	91.36%	**96.99%**
`continuity`	0.1244	**0.2793**
`neighborhood_hit`	28.78%	**93.67%**
`average_local_error`	46.06%	**22.17%**